

1. Louise is mixing red and blue paint colors to produce a shade of violet. The diagram represents 3 batches of the shade of violet that Louise created by mixing red and blue paint.

red paint (cups)



blue paint (cups)



- a. Select the phrases that correctly complete the sentence.

The ratio of ☐ red paint
☐ blue paint to ☐ red paint
☐ blue paint in Louise's mixture is $\frac{4}{2}$.

- b. Determine the value that correctly completes the sentence. Show your work in the space provided.

There are _____ cups of blue paint used for every 36 cups of violet that is mixed.

2. On June 22, 1947 in Holt, Missouri it rained 12 inches in 42 minutes. Complete the ratio to express the rate it rained on that day. Show your work in the space provided.

_____ minutes : 1 inch

3. Xiao Xing worked a 6-hour day and earned \$72.00. The double number line represents the amount she earned for working different numbers of hours.

wages earned
(dollars)



time worked
(hours)



Determine the amount of money Xiao Xing earns in 1 hour. Show your work in the space provided or on the double number line.

Xiao Xing earns \$_____ in 1 hour.

4. Marcelo installed solar panels on his roof. The panels produce 1,450 watt-hours when exposed to 5 hours of sunshine. Use the table to determine how long it will take the panels to produce 2,320 watts-hours at this rate. Show how the values in each row are determined using arrows and multiplication. Use however many rows you need to show your work. Then, complete the statement.

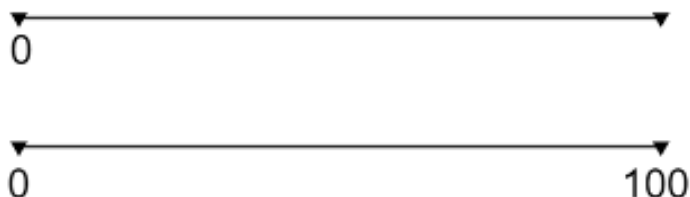
Hours	Electricity Produced (watts-hours)
5	1450

It will take _____ hours to produce 2,320 watts-hours.

5. A particular cleaning solution calls for $\frac{1}{4}$ cup of cleaning vinegar to be mixed with 2 cups of warm water. Complete the table to find the quantities needed for different batches of the cleaning solution.

Cleaning Vinegar (cups)	Warm Water (cups)	Cleaning Solution (cups)
$\frac{1}{4}$	2	
	16	
		27

6. Determine what percentage of 120 is 72. Show your reasoning on the double number line diagram. Then, complete the statement.



72 is ____% of 120.

7. Marci hikes 15 miles in 6 hours on the Florida Trail.
- a. Find the unit rate. Include units.

b. Interpret the unit rate for Marci's hike.

8. A sorting machine at an orange processing plant has to sort 8,640 tons of oranges to meet a deadline. The machine can process 36,000 tons of oranges in 50 minutes. Determine how long will it take the machine to sort 8,640 tons of oranges. Show your work in the space provided. Then, complete the statement.

It will take _____

- ☐ hours.
- ☐ minutes.
- ☐ tons.

Work

9. At a clothing store, a pair of jeans costs \$40. Using a coupon, Shanice is able to pay \$10 less than the original price.

Determine the percentage off the original price the coupon represents. Show your work in the space provided. Then, complete the statement.

The coupon gives Shanice ____% off of the original price.

Work

10. Use the conversion rates to find each value. Show your work.

358 centimeters = _____ meters	Work
7 gallons = _____ pints	Work

Unit Conversions in the Metric System

1 meter (m) = 100 centimeters (cm)

1 meter (m) = 1,000 millimeters (mm)

1 kilometer (km) = 1,000 meters (m)

Unit Conversions in the Customary System

1 cup (c) = 8 fluid ounces (fl oz)

1 pint (pt) = 2 cups (c)

1 quart (qt) = 2 pints (pt)

1 gallon (gal) = 4 quarts (qt)